

iNKT: cell states, cytotoxic programs and functional changes

Integrated editable presentation | main talk + complete source-page appendix

Research question

How does the T2-associated response differ across tissues and iNKT states?

Presentation sequence

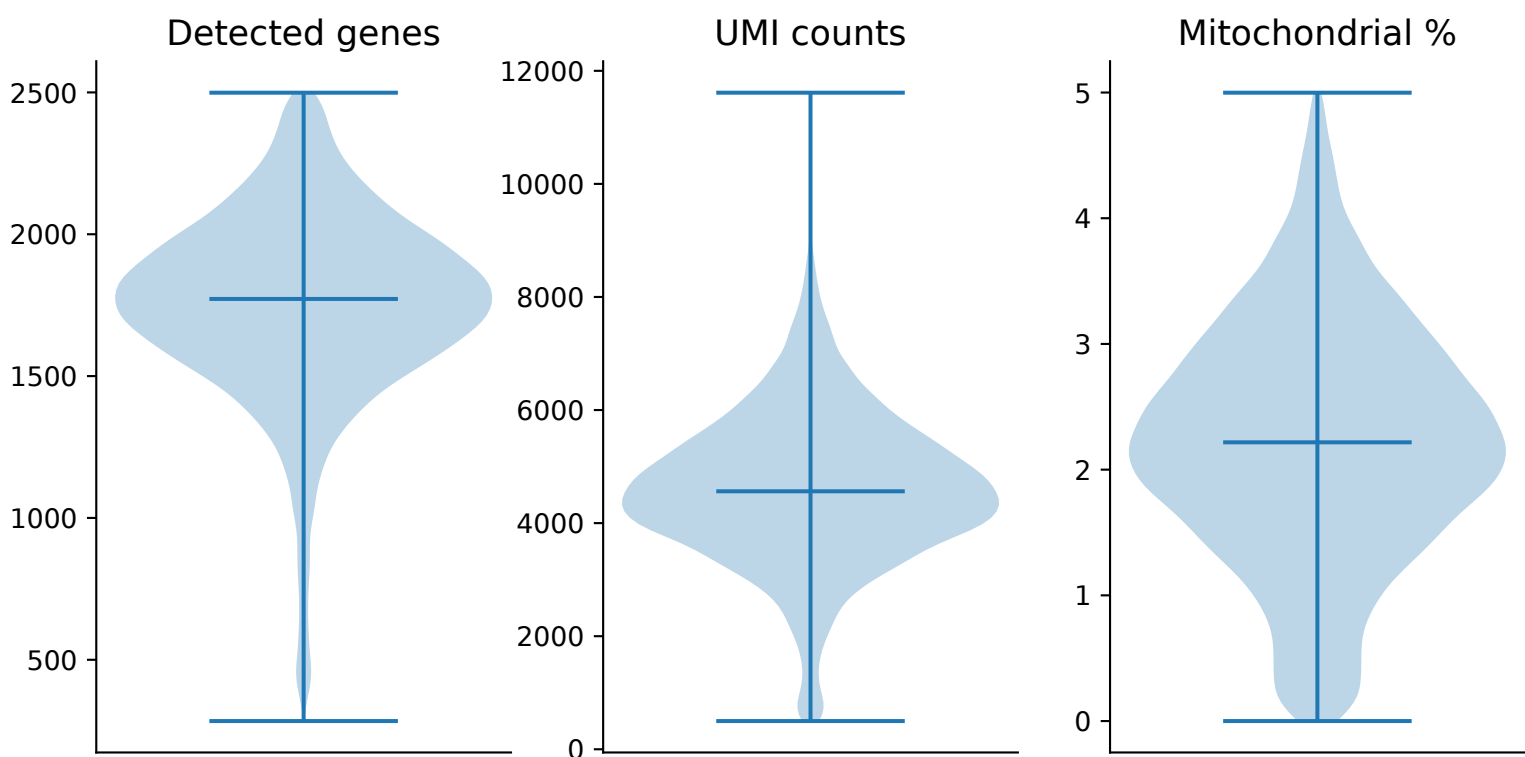
- 1 Cohort, cell-state markers and tissue composition
- 2 Differential expression, cytotoxic modules and GO functions
- 3 Local state ordering, robustness and bounded interpretation

15,532 cells | 3 tissues | Ctrl / T2 | 9 original refined clusters
One sample label per tissue x condition: all biological conclusions remain exploratory.

Material, dataset and QC

Six samples • 15,532 retained cells • 10,670 measured genes

Sample	Input cells	Retained
Ctrl_Thymus	2101	1372
Ctrl_BM	3998	3379
Ctrl_Spleen	3829	3422
T2_Thymus	1639	1257
T2_BM	3131	2731
T2_Spleen	3760	3371

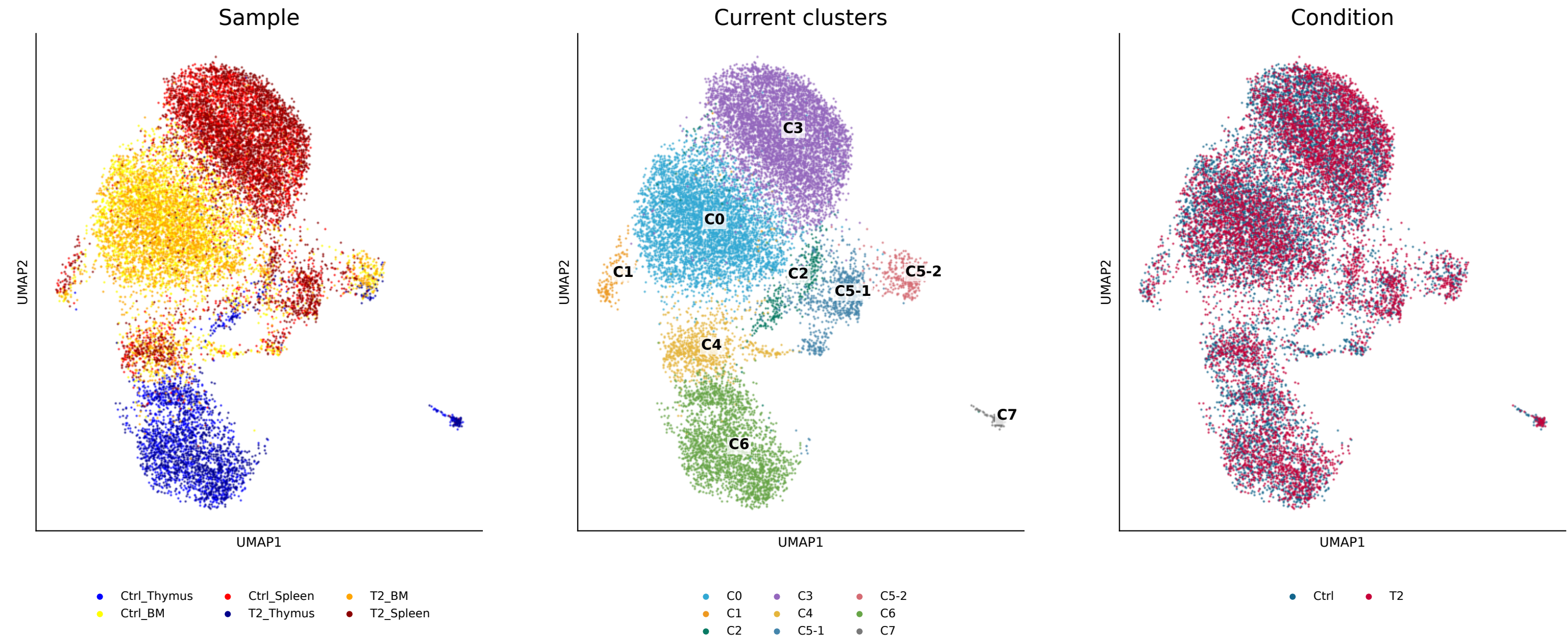


Gene min_cells = 100, then
 $200 \leq \text{detected genes} < 2,500$; mitochondrial fraction $< 5\%$.

QC counts match the legacy deck; downstream UMAP, Leiden and gene statistics remain a reanalysis.

Clusters and tissue composition

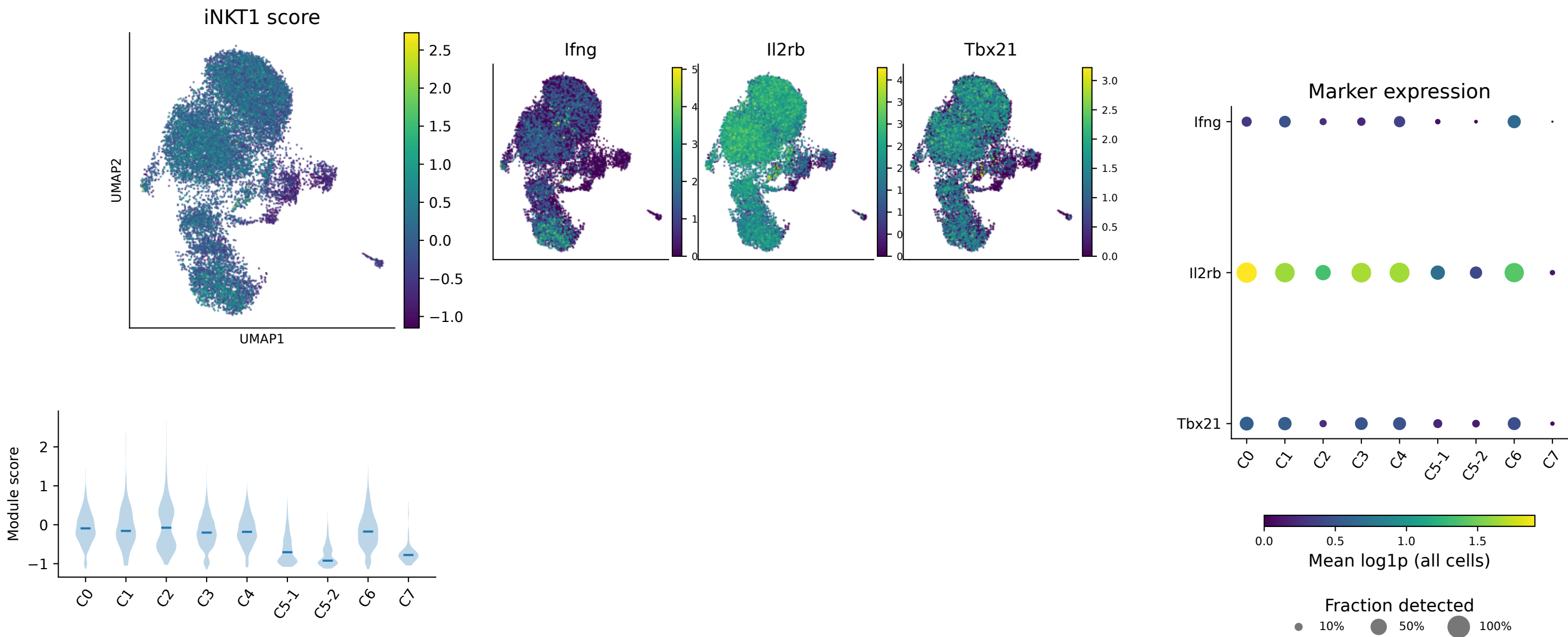
Fixed 50-PC UMAP; C5 refined to C5-1 and C5-2



Original sample hues retained on the left; Blood-style blue/red encodes condition on the right. Coordinates are unchanged.

iNKT1 marker program

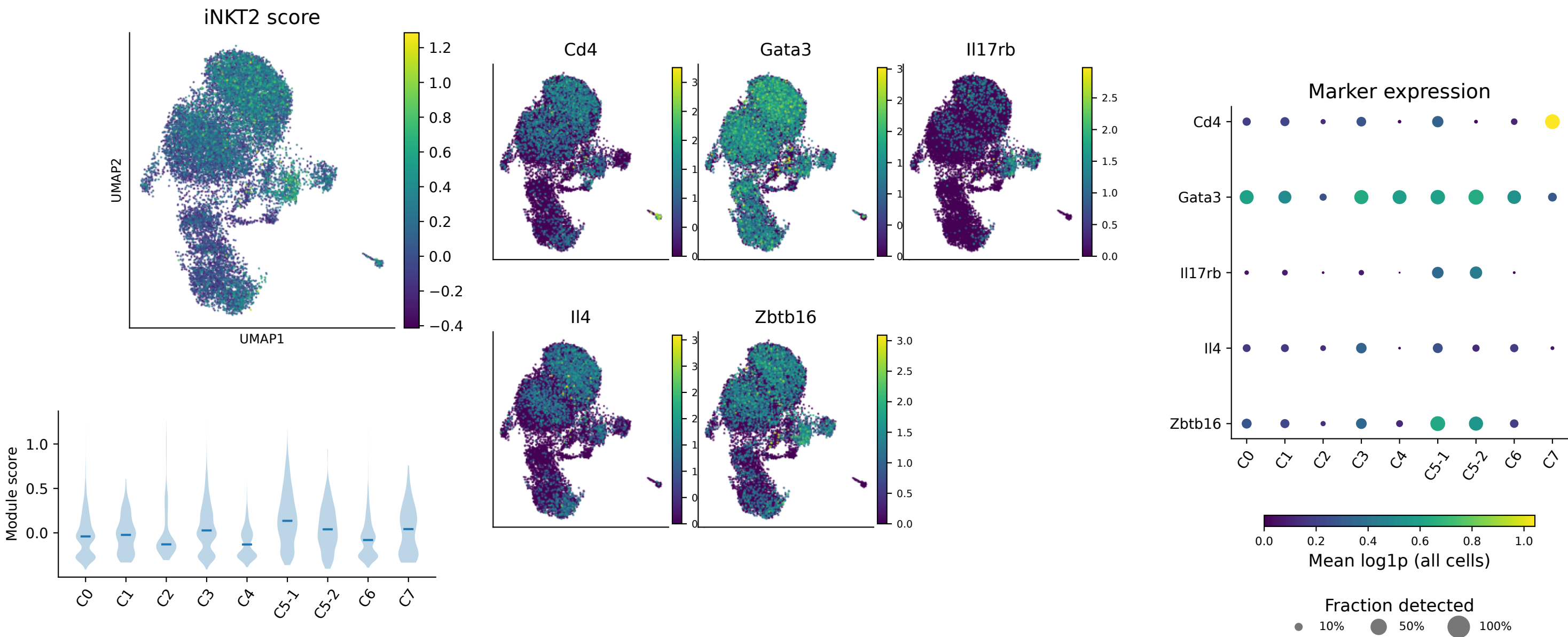
Score map, cluster distributions, individual markers and expression dotplot



Current score_genes units differ from legacy summed-marker scores; these panels do not establish discrete subtype identities.

iNKT2 marker program

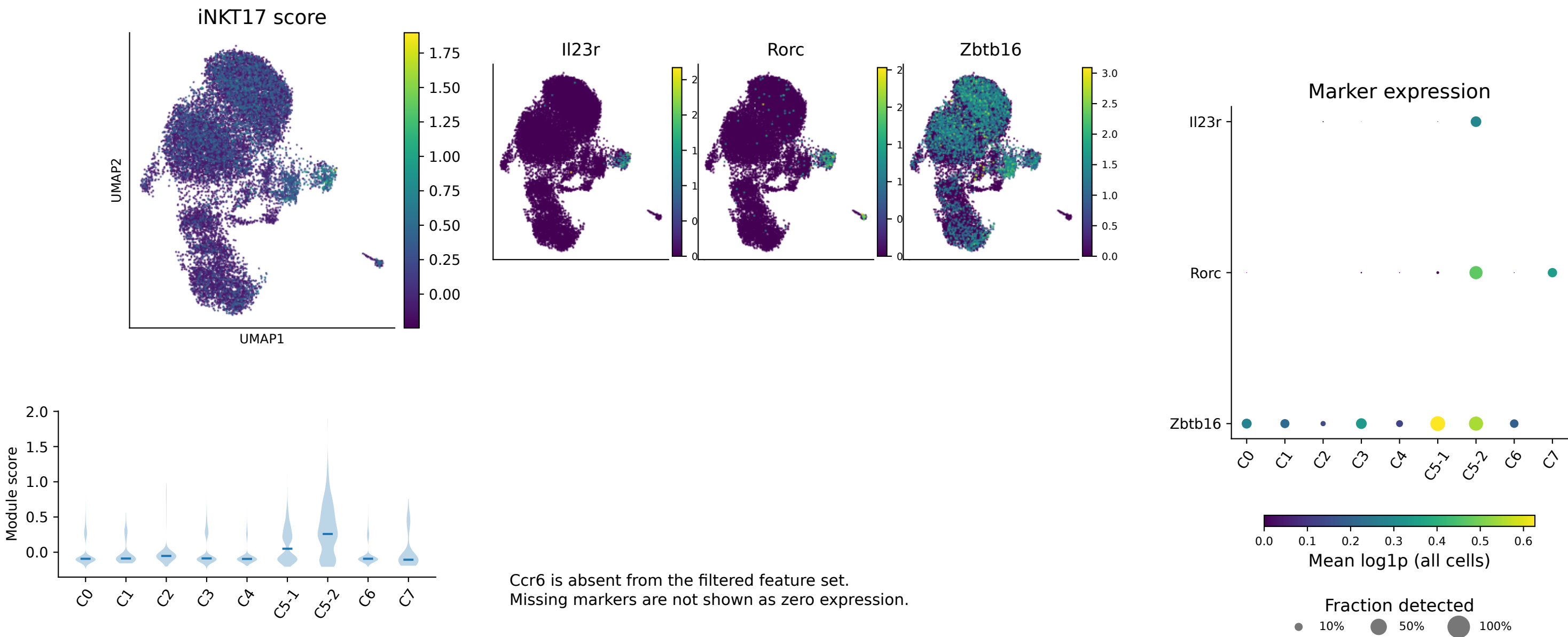
Score map, cluster distributions, individual markers and expression dotplot



Current score_genes units differ from legacy summed-marker scores; these panels do not establish discrete subtype identities.

iNKT17 marker program

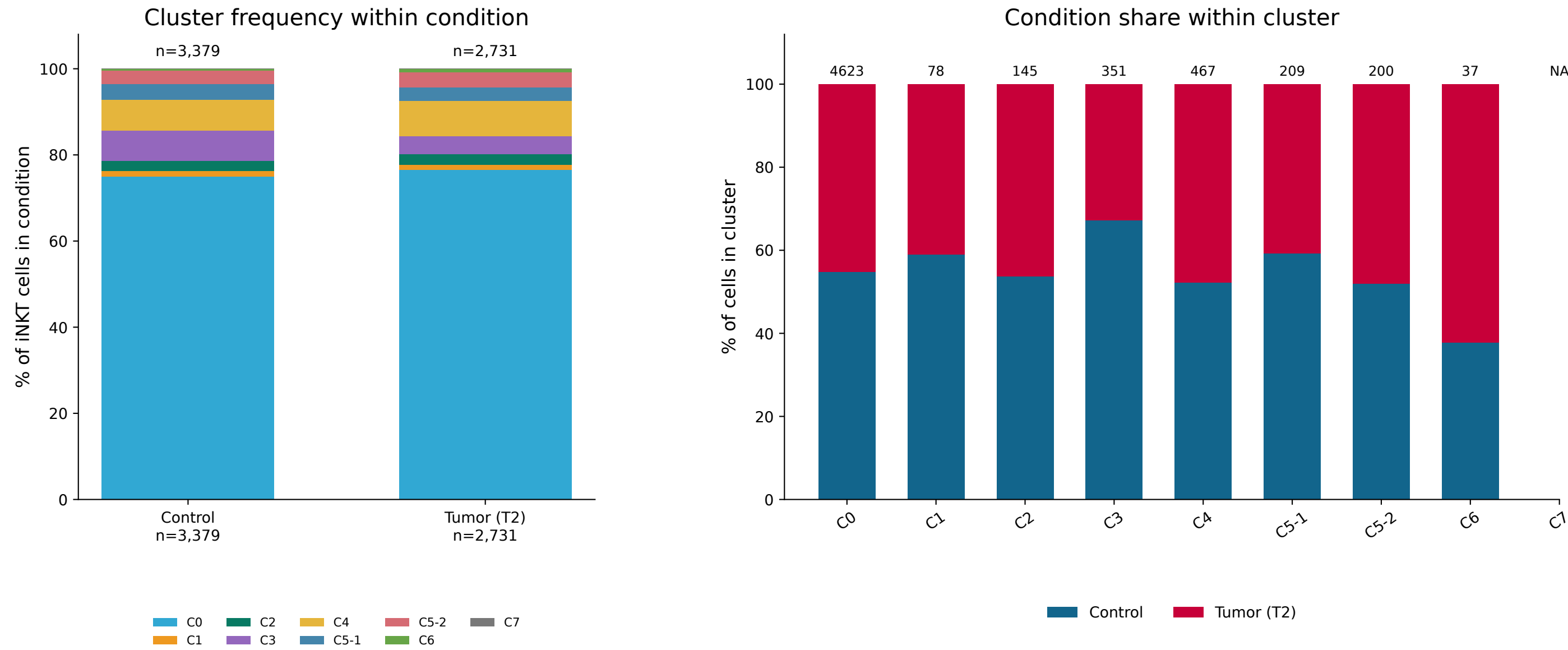
Score map, cluster distributions, individual markers and expression dotplot



Current score_genes units differ from legacy summed-marker scores; these panels do not establish discrete subtype identities.

Bone marrow: cluster and condition frequencies

C • The two denominators from Blood Figure 3C, presented side by side

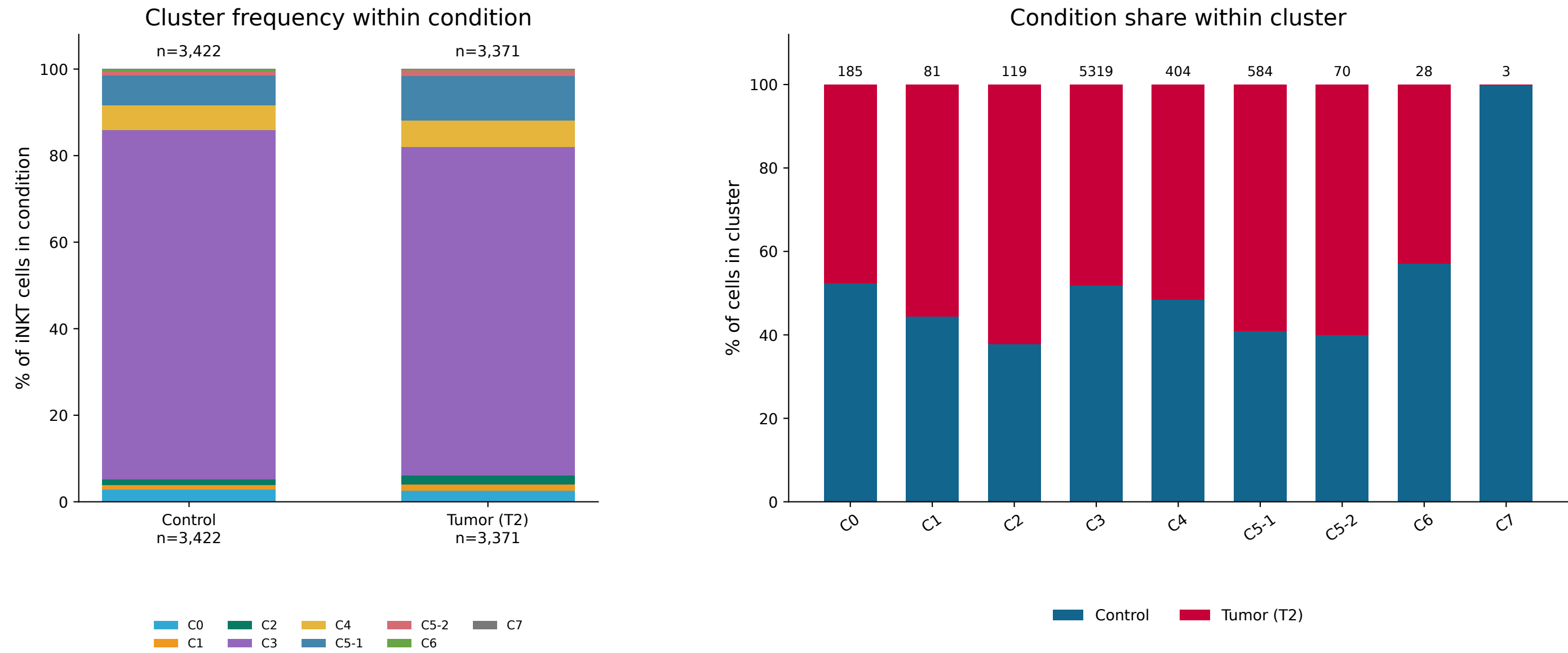


Left: $n(\text{cluster, condition}) / N(\text{condition})$ | Right: $n(\text{condition, cluster}) / N(\text{cluster})$

Captured-cell proportions only. No mouse-level error bars or significance are inferred from individual cells.

Spleen: cluster and condition frequencies

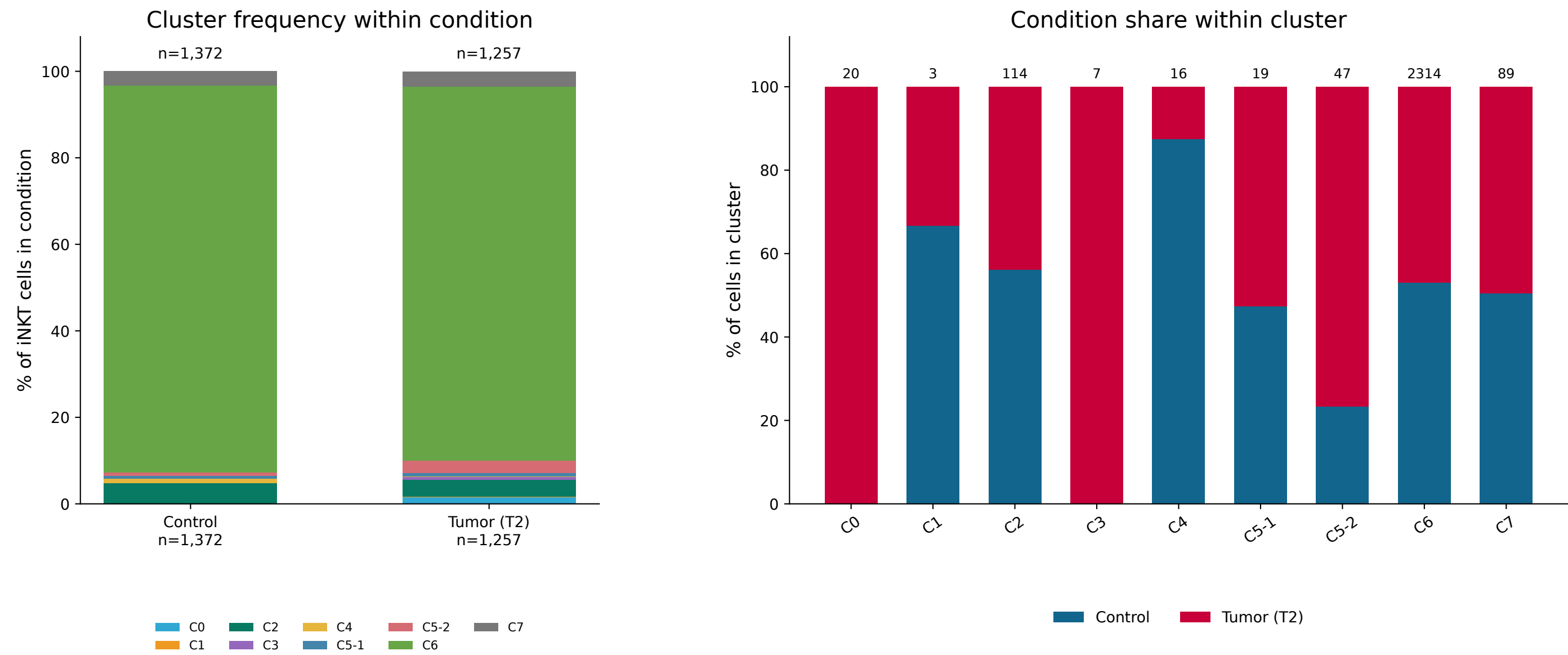
C • The two denominators from Blood Figure 3C, presented side by side



Captured-cell proportions only. No mouse-level error bars or significance are inferred from individual cells.

Thymus: cluster and condition frequencies

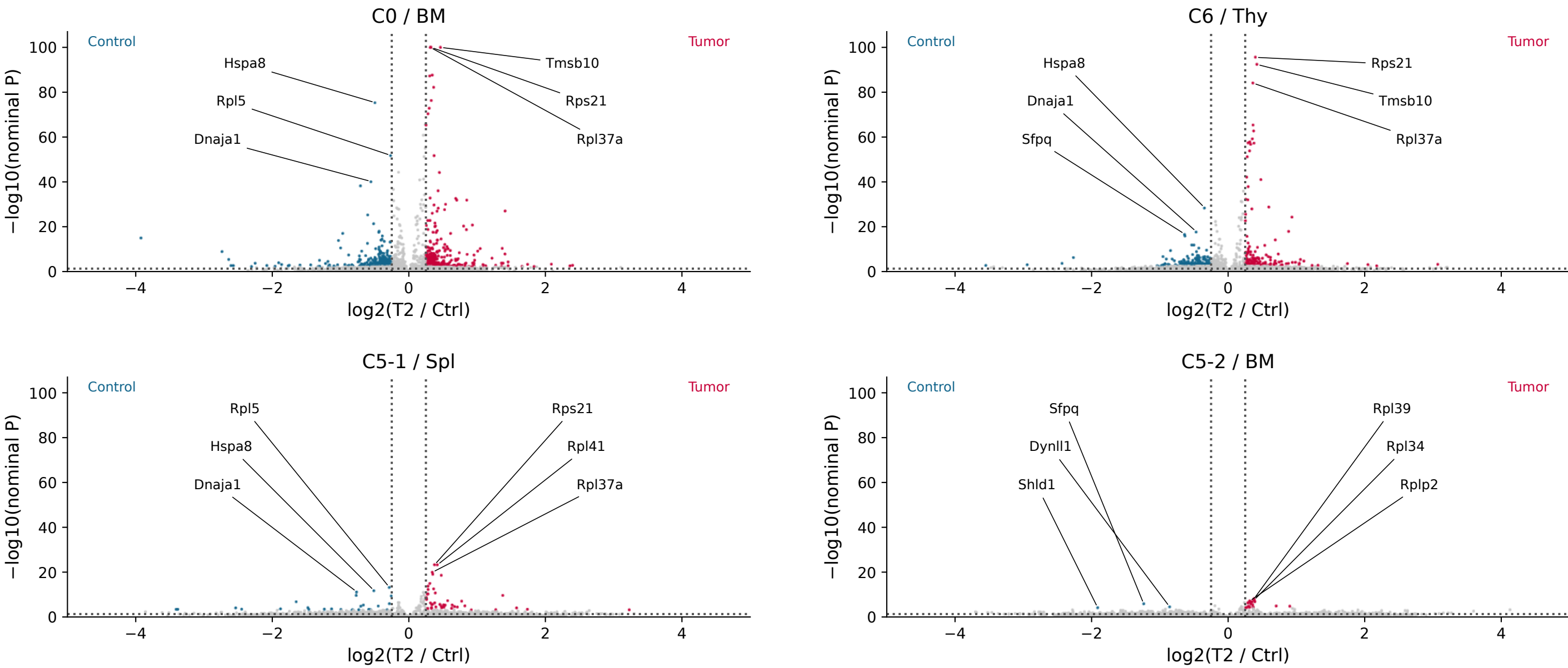
C • The two denominators from Blood Figure 3C, presented side by side



Captured-cell proportions only. No mouse-level error bars or significance are inferred from individual cells.

D Differential expression: Control vs. Tumor

Paper axes and blue/red direction; current tissue × cluster comparisons



Common axes: $|\log_2\text{FC}| \leq 5$; $-\log_{10}(P)$ capped at 100. All uncapped values and display flags are exported.

Color rule retained: gene BH FDR ≤ 0.05 and $|\log_2\text{FC}| \geq 0.25$. Vertical display thresholds are therefore ± 0.25 .

Analysis design and reference mapping

Methods adapted to mouse iNKT; biological conclusions are derived from these data.

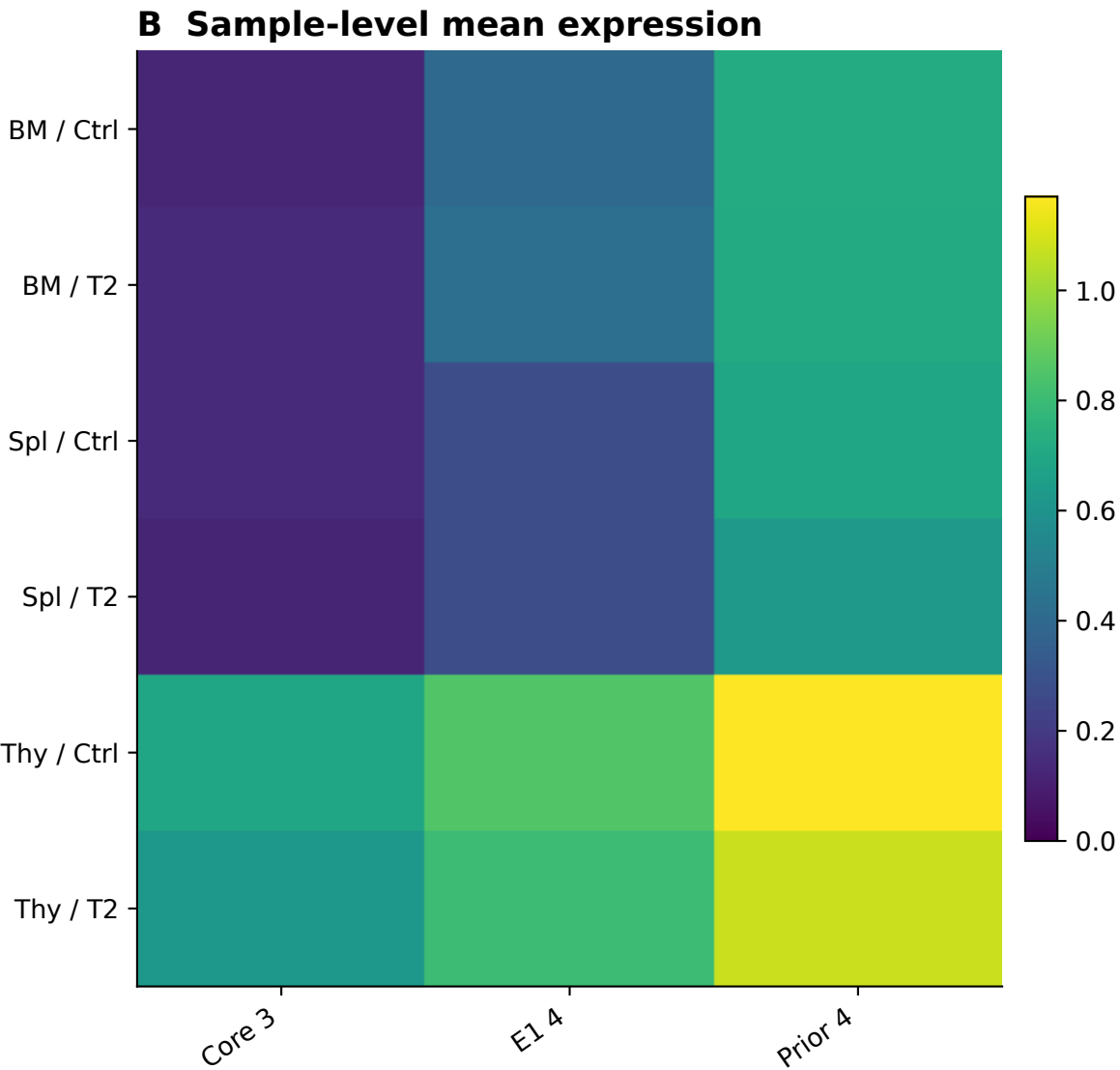
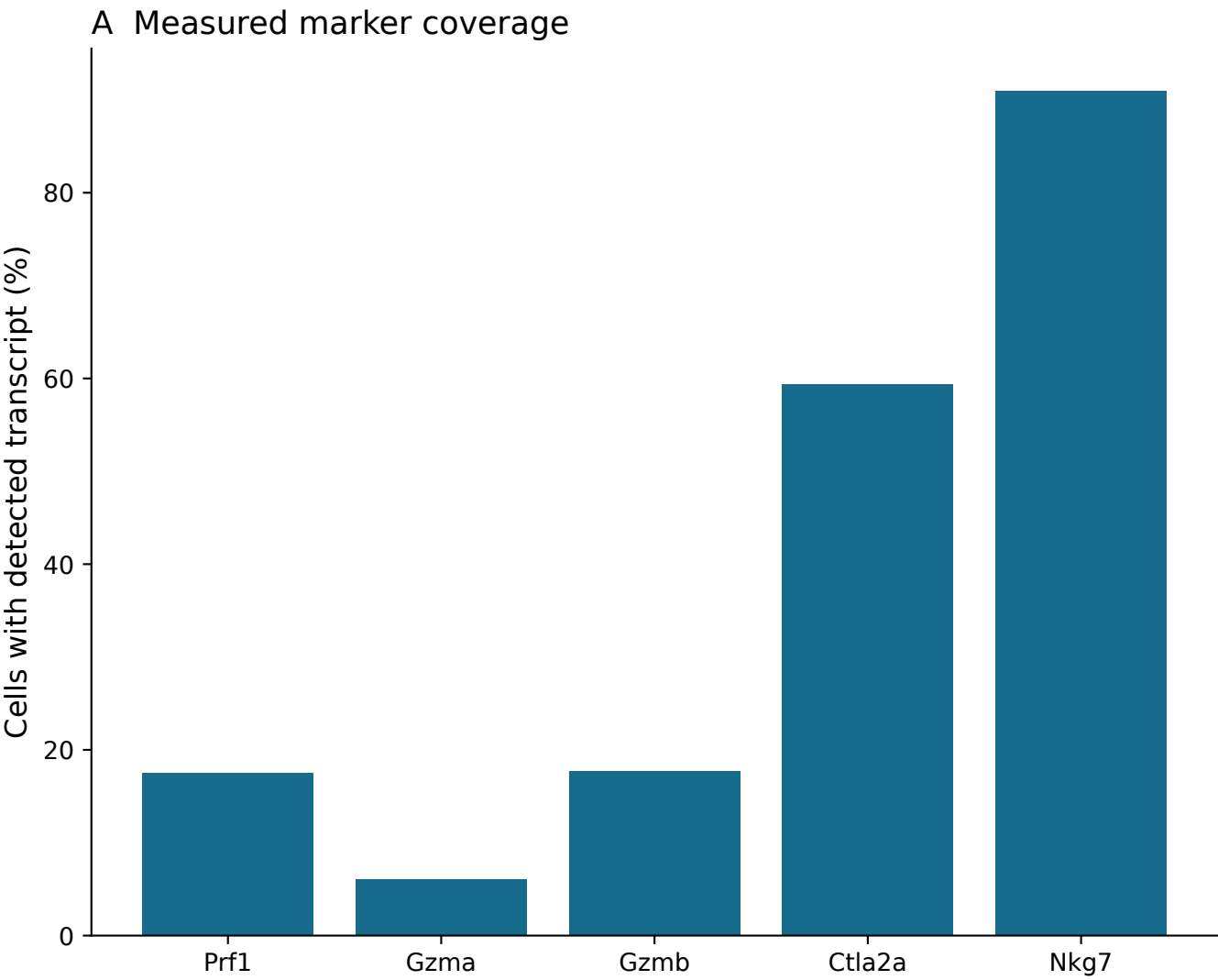
Analysis	Reference	Input / definition	Implementation
Cytotoxicity	PAGER Fig. 5C/D	Core: Prf1, Gzma, Gzmb; E1: +Ctla2a	Mean log1p expression; matched-control score sensitivity
GO cluster identity	GAFA Fig. 4c	Cluster vs rest within each tissue	Wilcoxon, gene BH <= .05 and log2FC >= .25
GO treatment response	GAFA functional comparison	T2-up and Ctrl-up genes tested separately	Hypergeometric; full eligible GO BP/MF/CC families
Local trajectories	PAGER Fig. 5D	Recompute diffusion map on tissue-specific graphs	Precursor-rooted DPT; 20 shared bins; 3-root sensitivity
Robustness	Adaptation to this dataset	Stable clusters; five QC-matched resamples	Effects and direction consistency; no animal replication

GO uses official mouse annotations. An enrichment of upregulated genes does not prove pathway activation.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Cytotoxic gene coverage and scoring

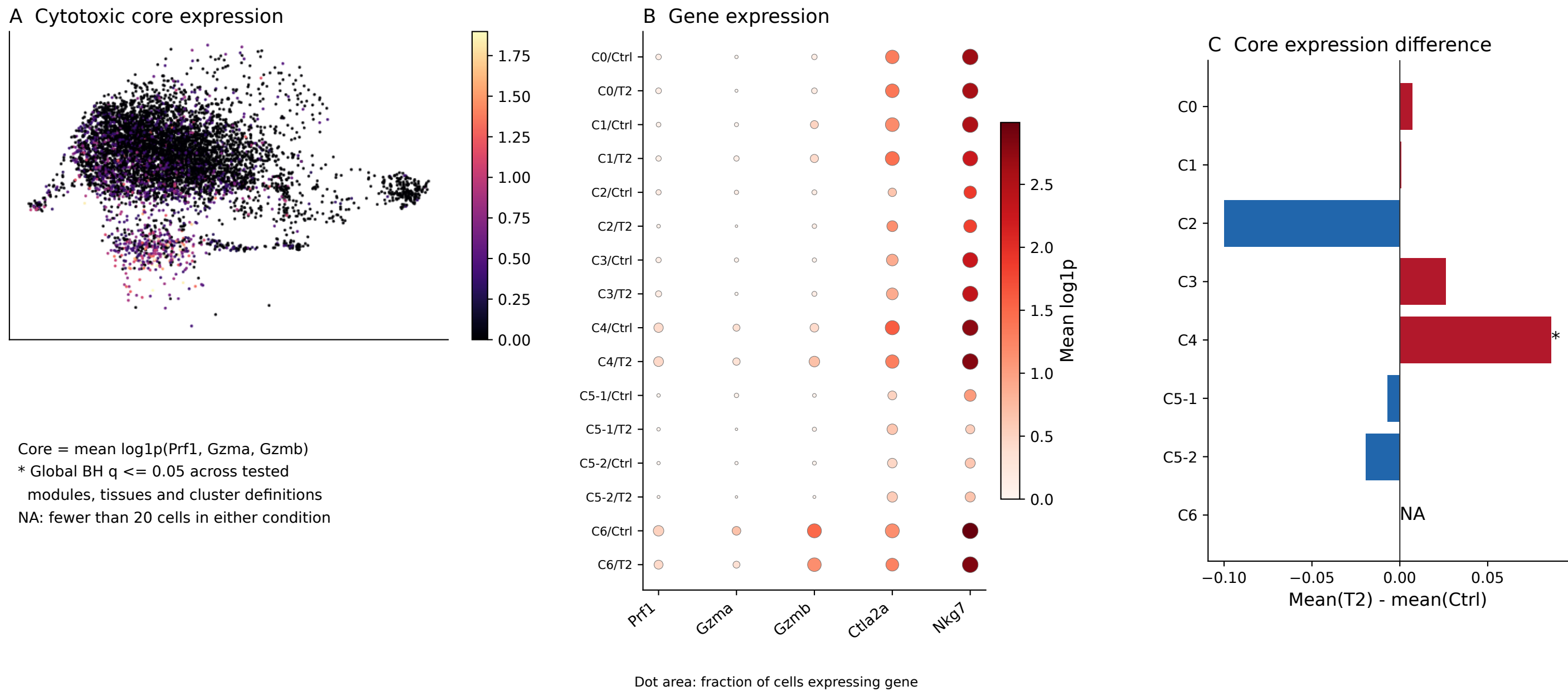
Primary endpoint: mean log-normalized expression of Prf1 / Gzma / Gzmb.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Bone marrow: cytotoxic expression and condition effects

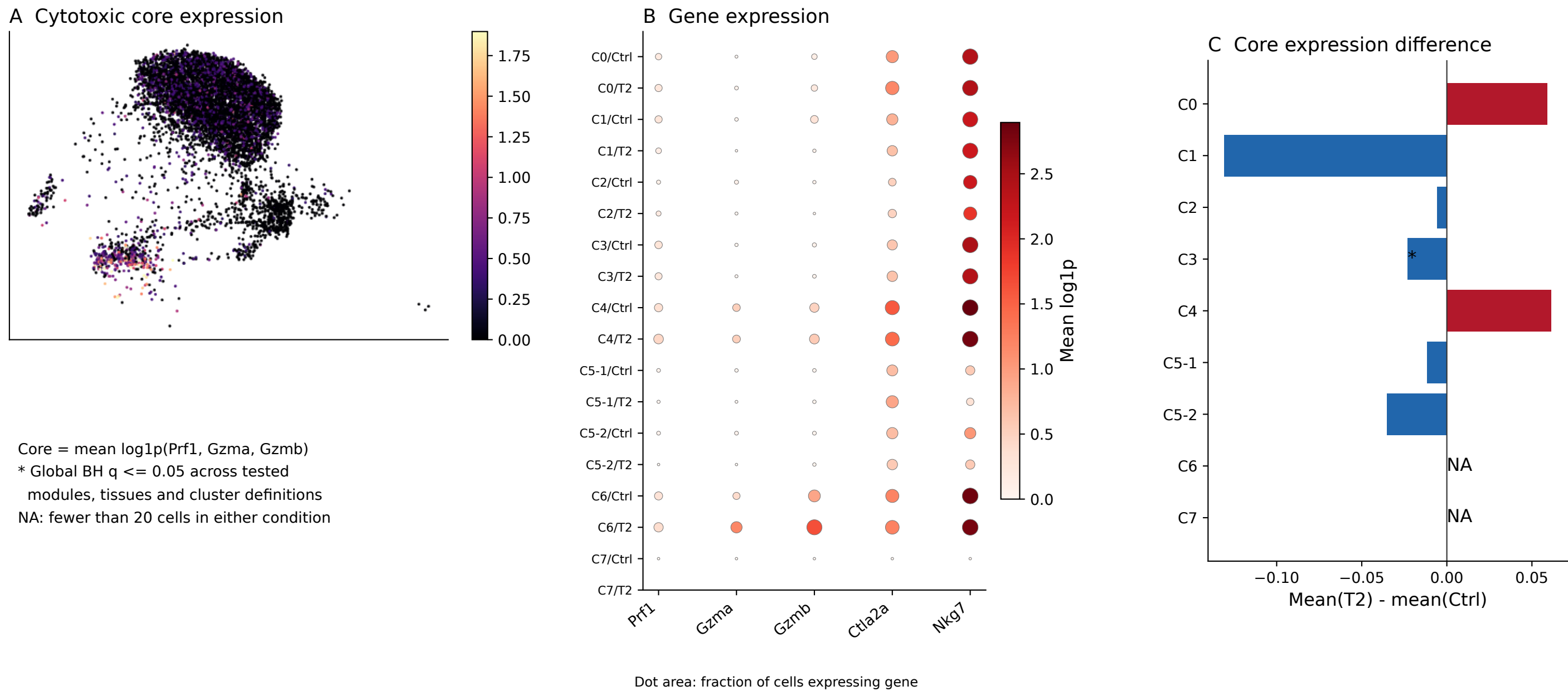
Original refined clusters; T2 - Ctrl within the same tissue and cluster.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Spleen: cytotoxic expression and condition effects

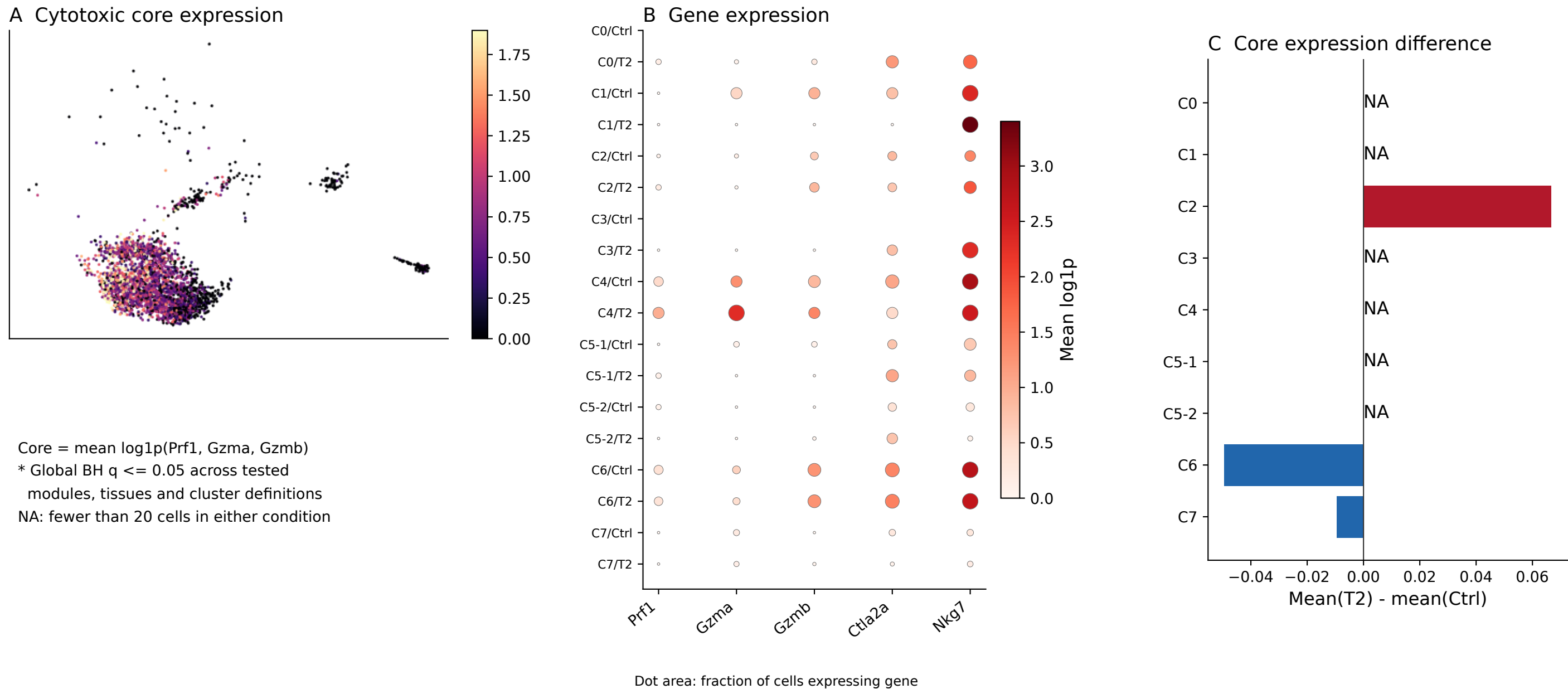
Original refined clusters; T2 - Ctrl within the same tissue and cluster.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Thymus: cytotoxic expression and condition effects

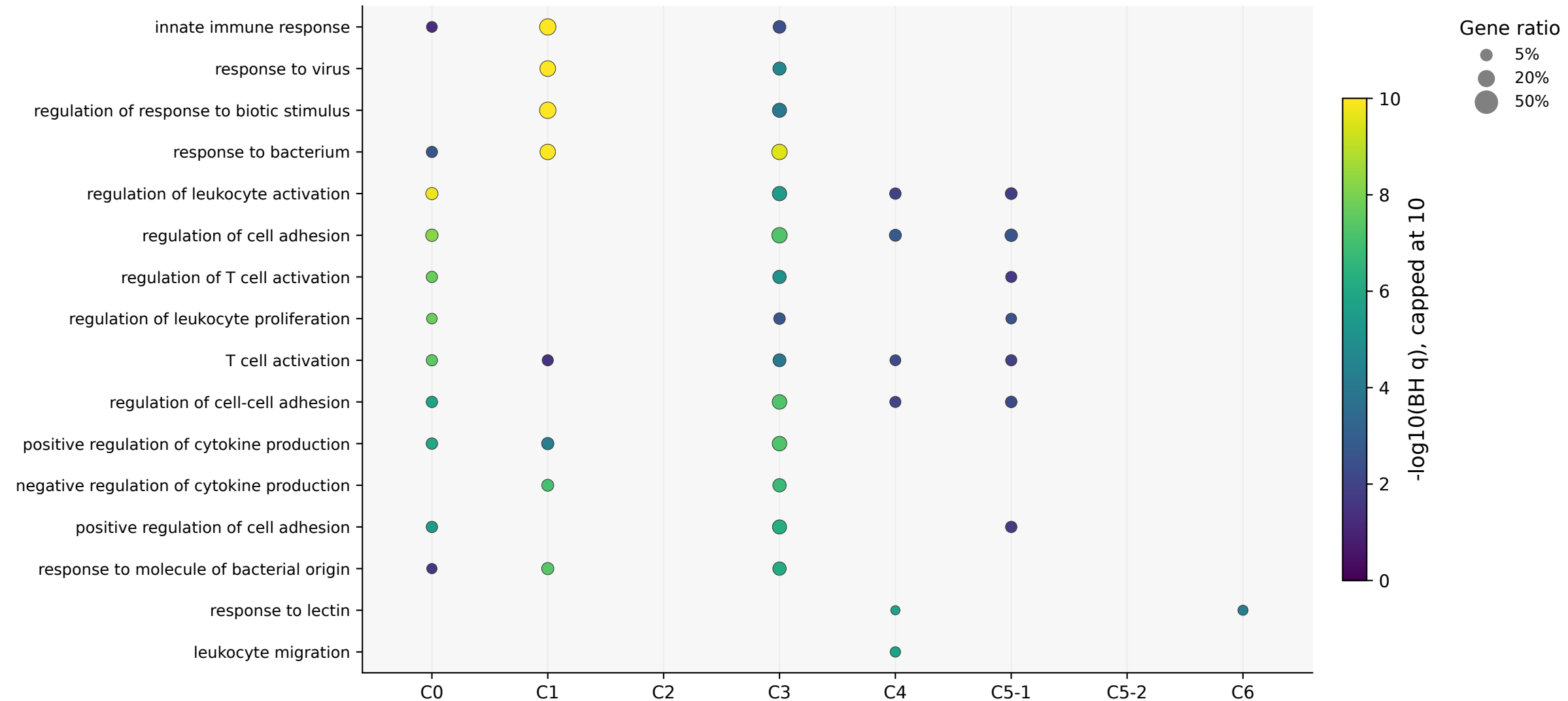
Original refined clusters; T2 - Ctrl within the same tissue and cluster.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Bone marrow: GO biological processes: cluster identity

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

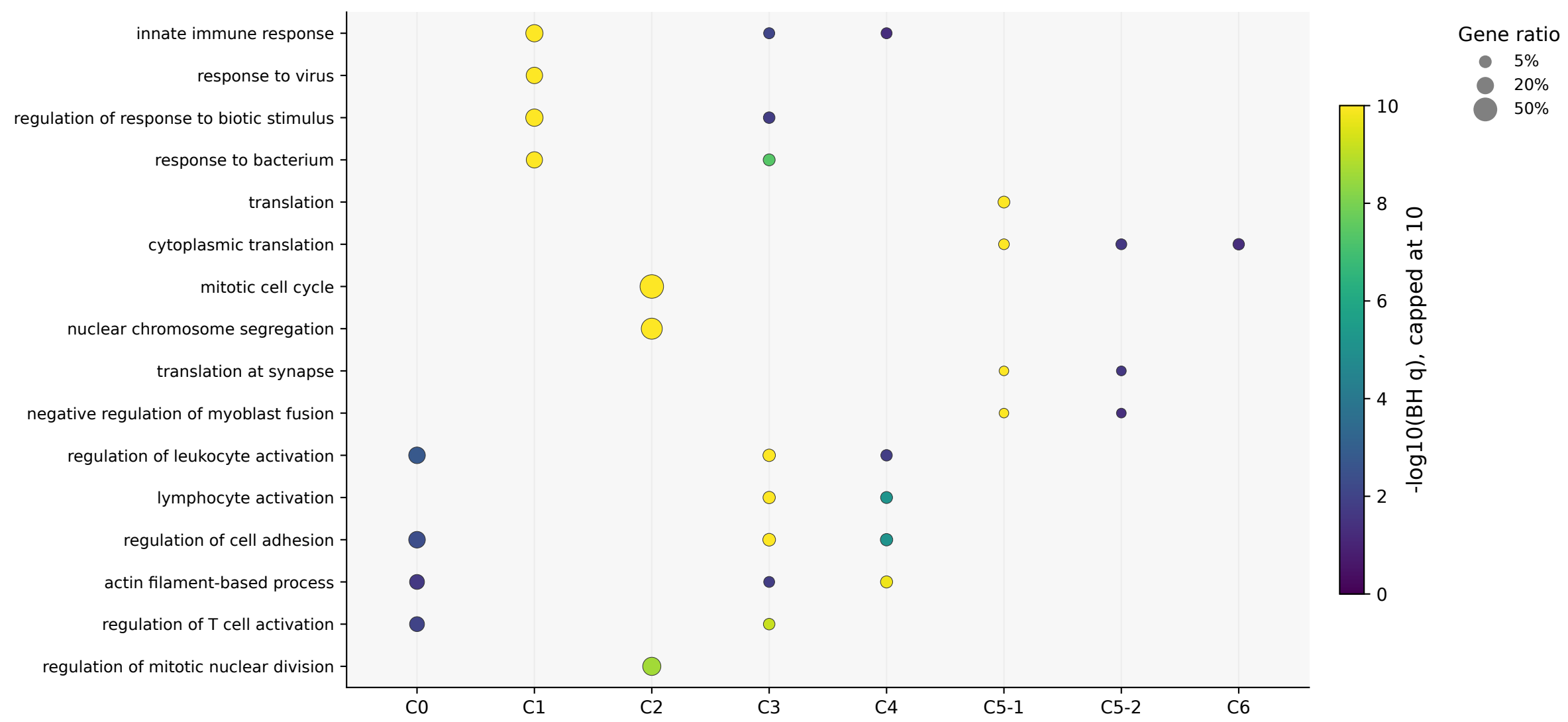


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; $q \leq .05$; ≥ 2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Spleen: GO biological processes: cluster identity

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

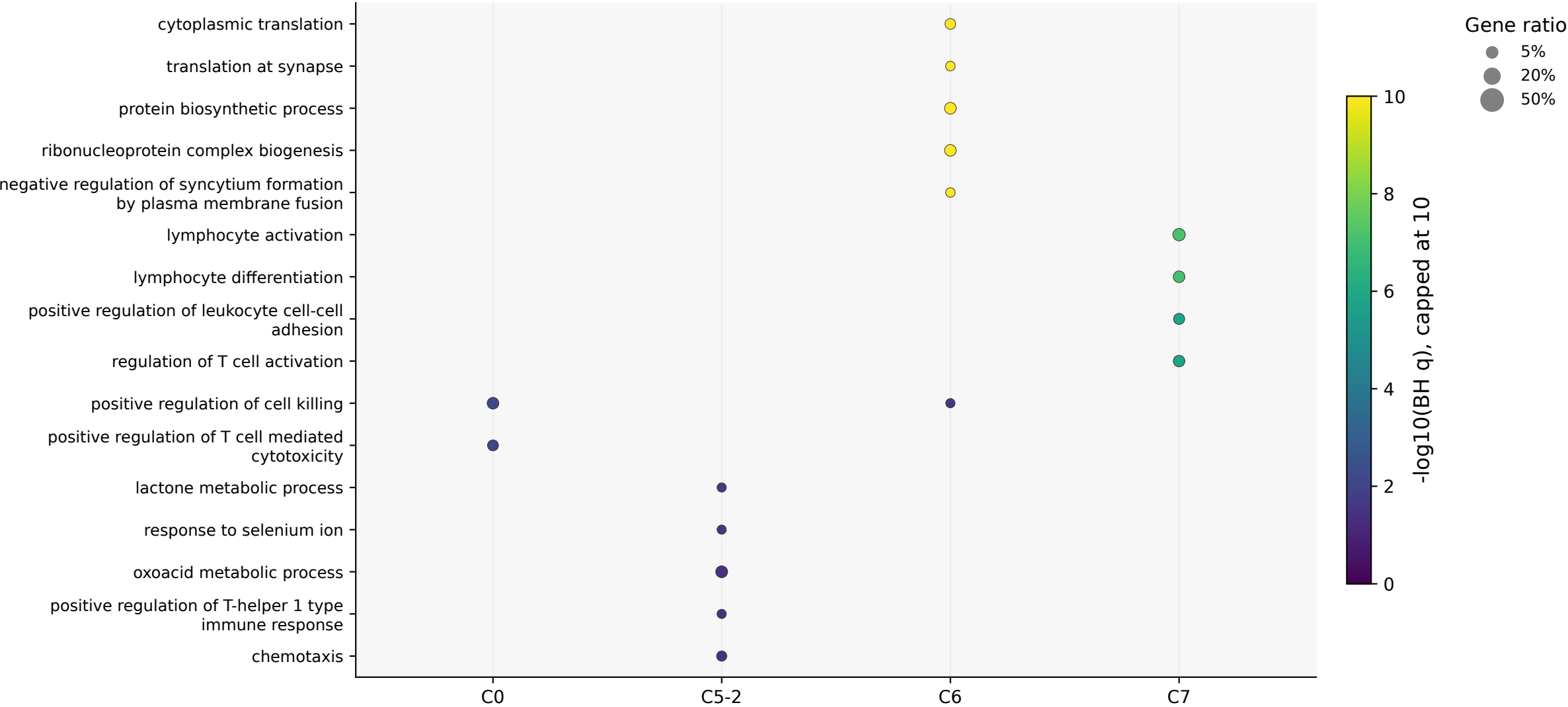


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; $q \leq .05$; ≥ 2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Thymus: GO biological processes: cluster identity

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

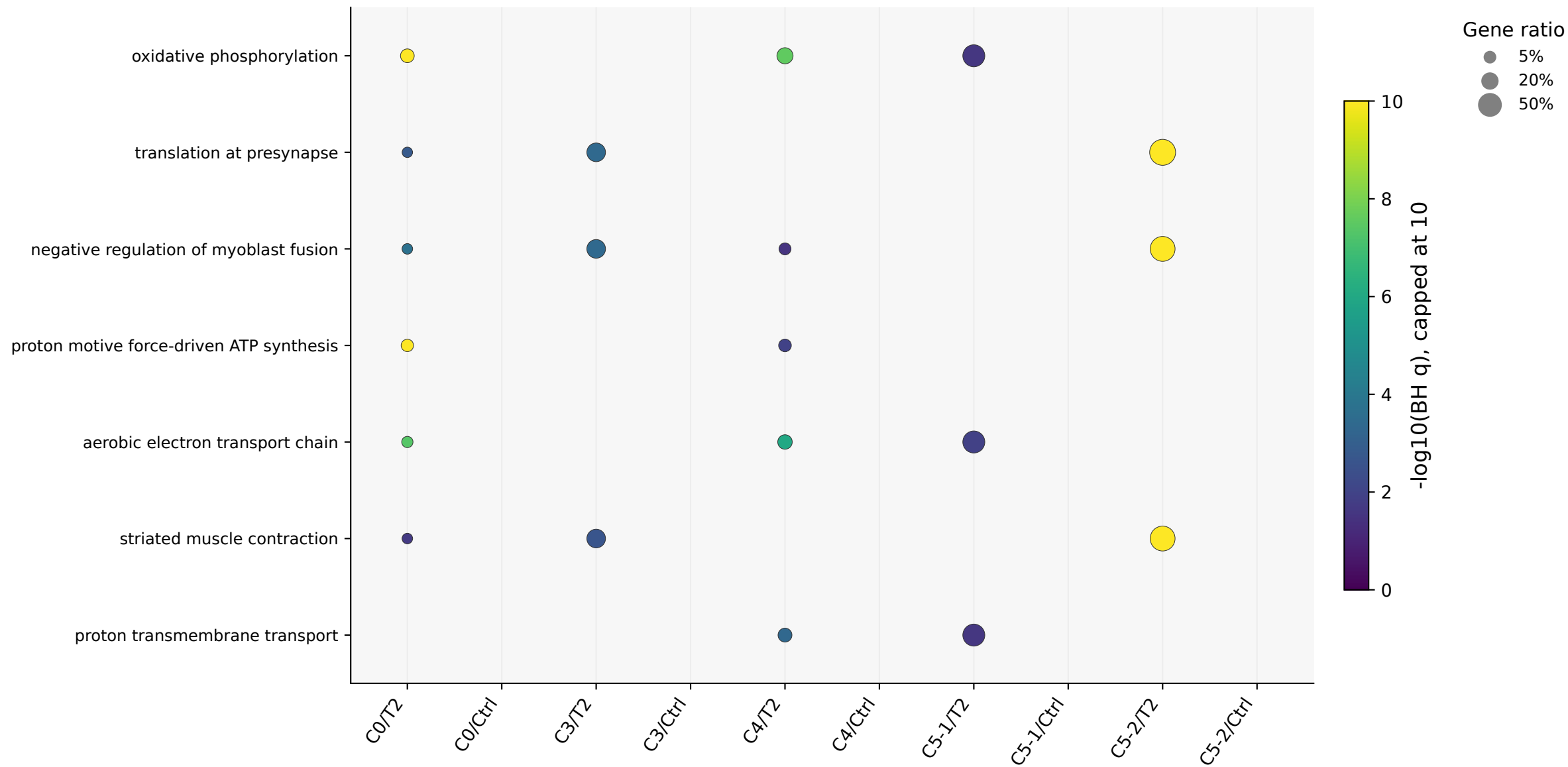


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; $q \leq .05$; ≥ 2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Bone marrow: GO biological processes: condition response

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

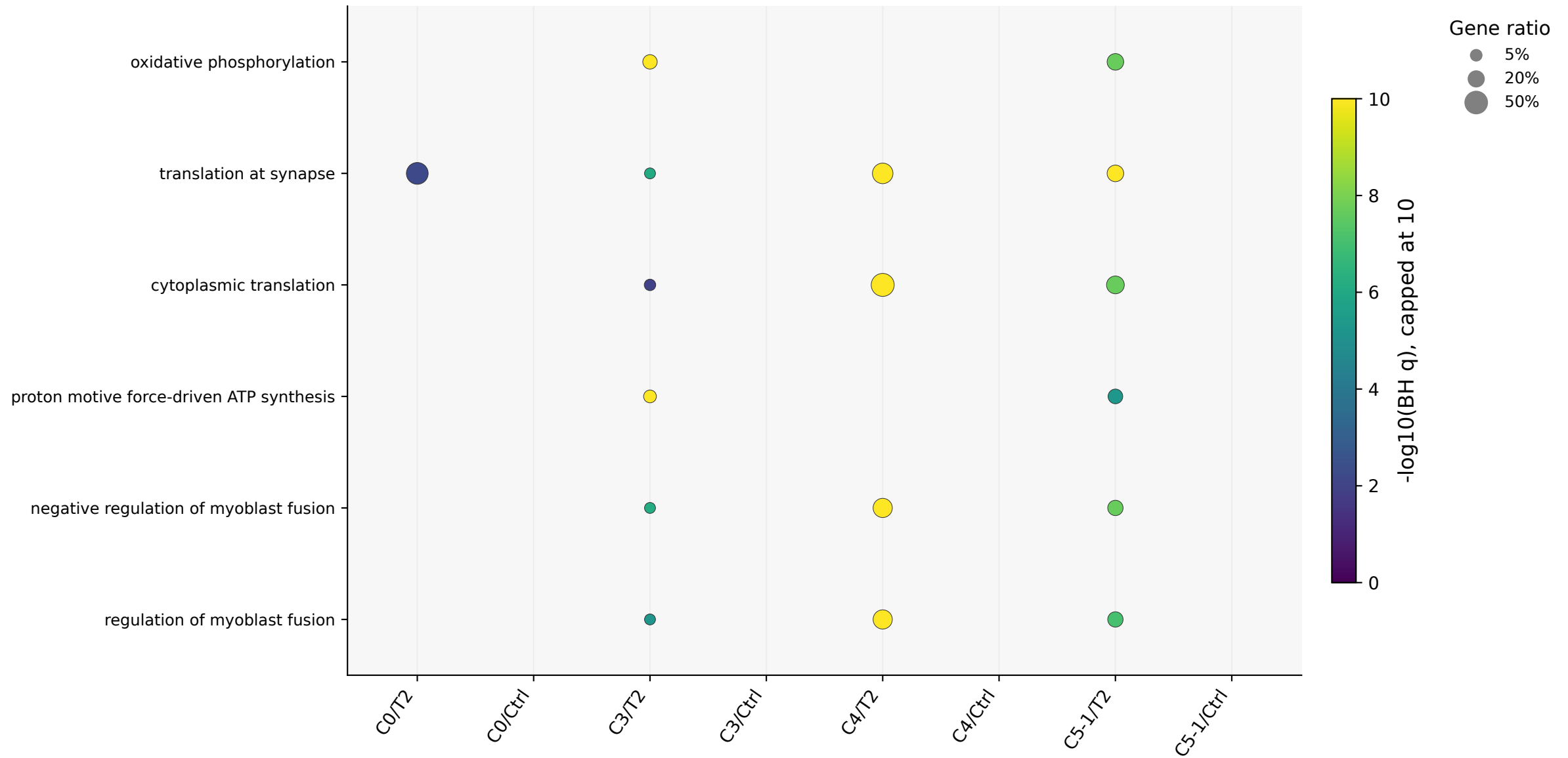


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; $q \leq .05$; ≥ 2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Spleen: GO biological processes: condition response

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

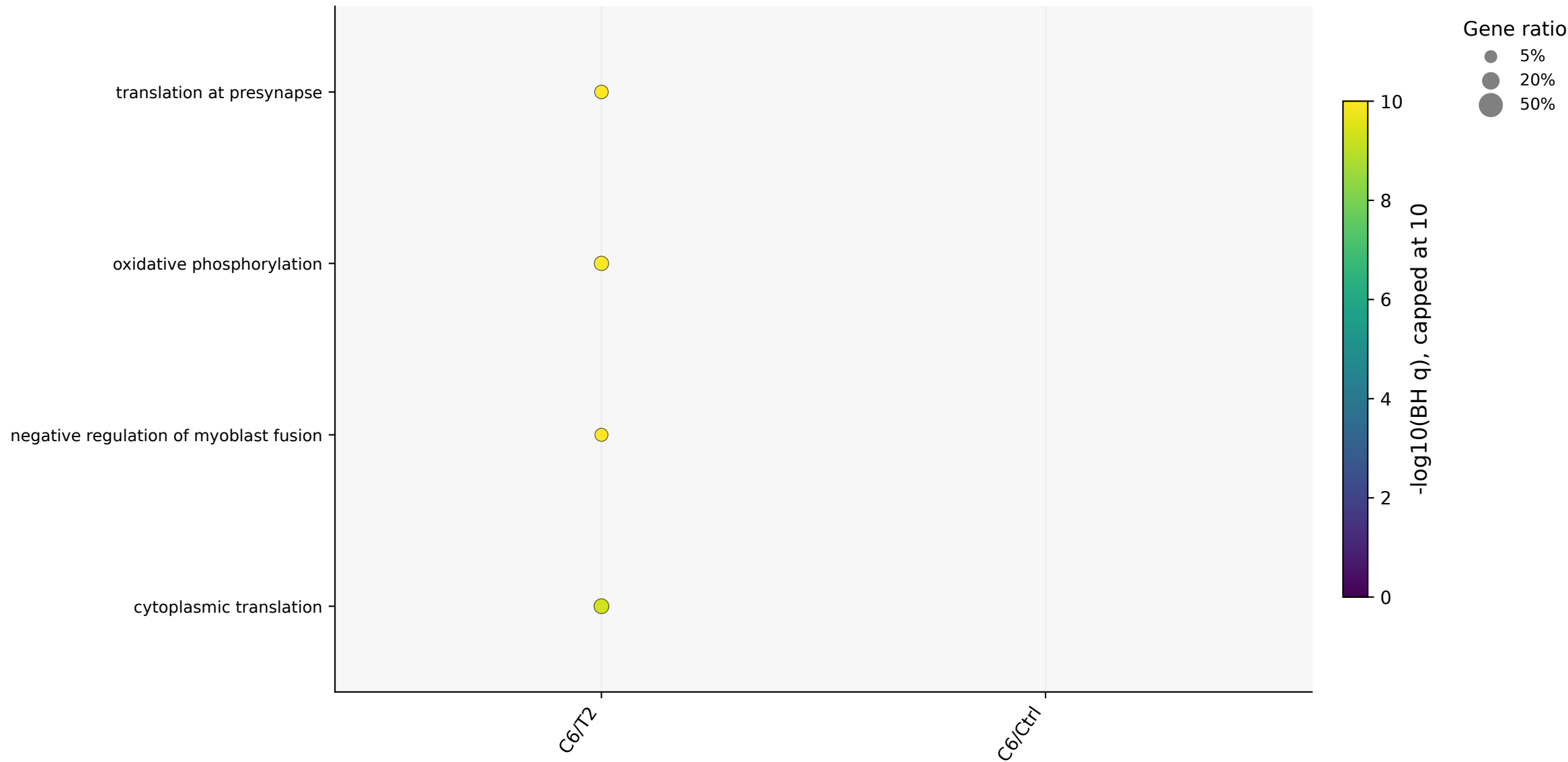


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; $q \leq .05$; ≥ 2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Thymus: GO biological processes: condition response

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

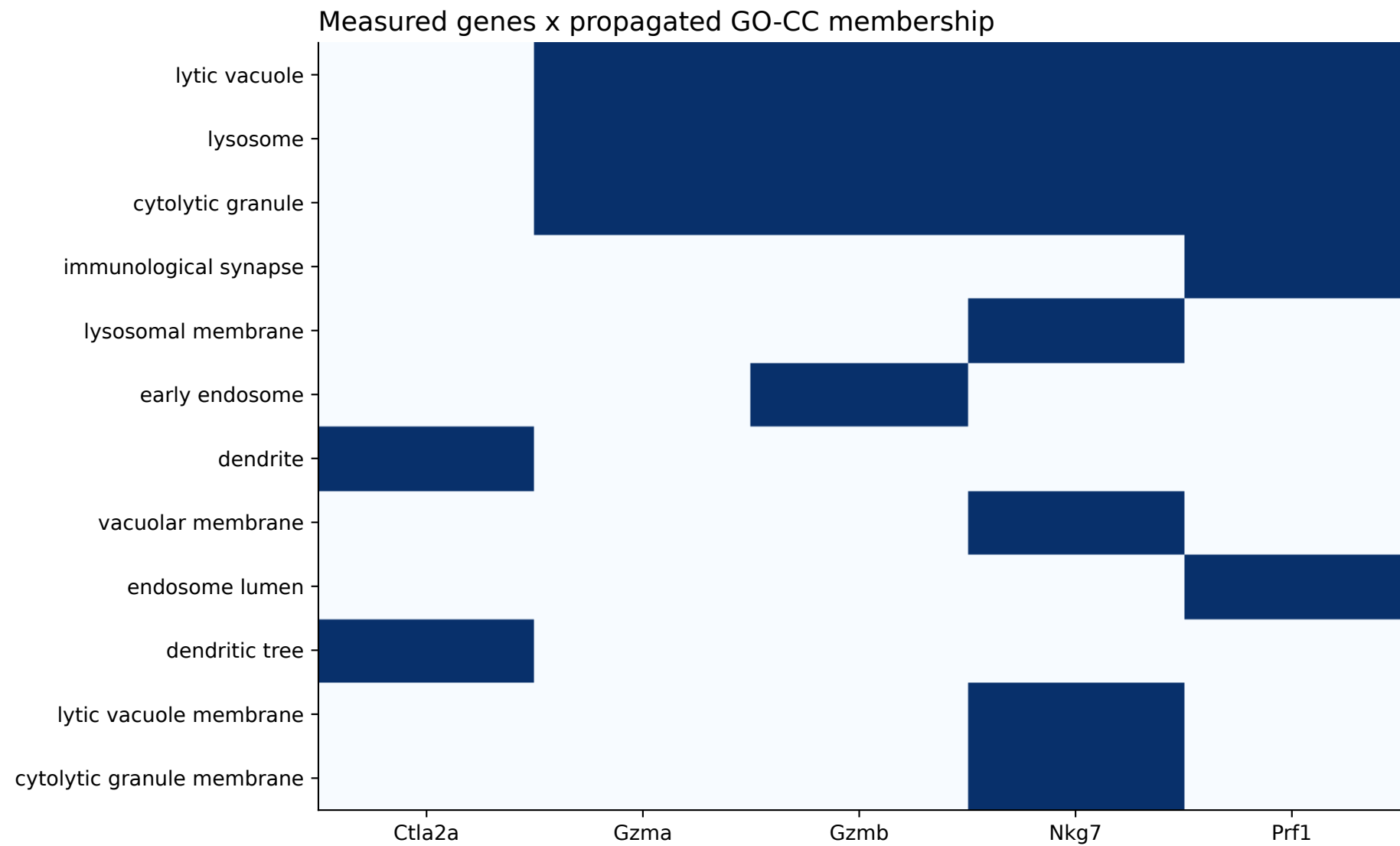


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; $q \leq .05$; ≥ 2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Cellular components of cytotoxic module genes

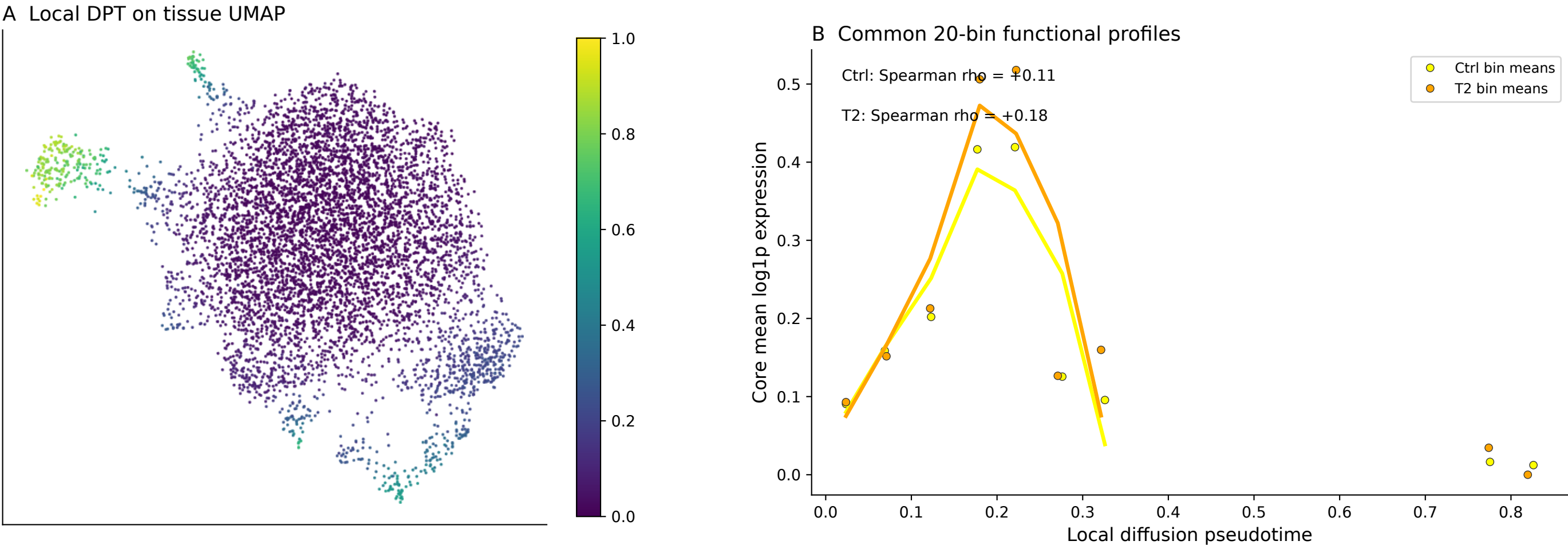
GO-CC membership supports localization; this is not a protein-interaction network.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Bone marrow: local cytotoxicity along pseudotime

PAGER-style functional curves on a newly computed within-tissue diffusion map.

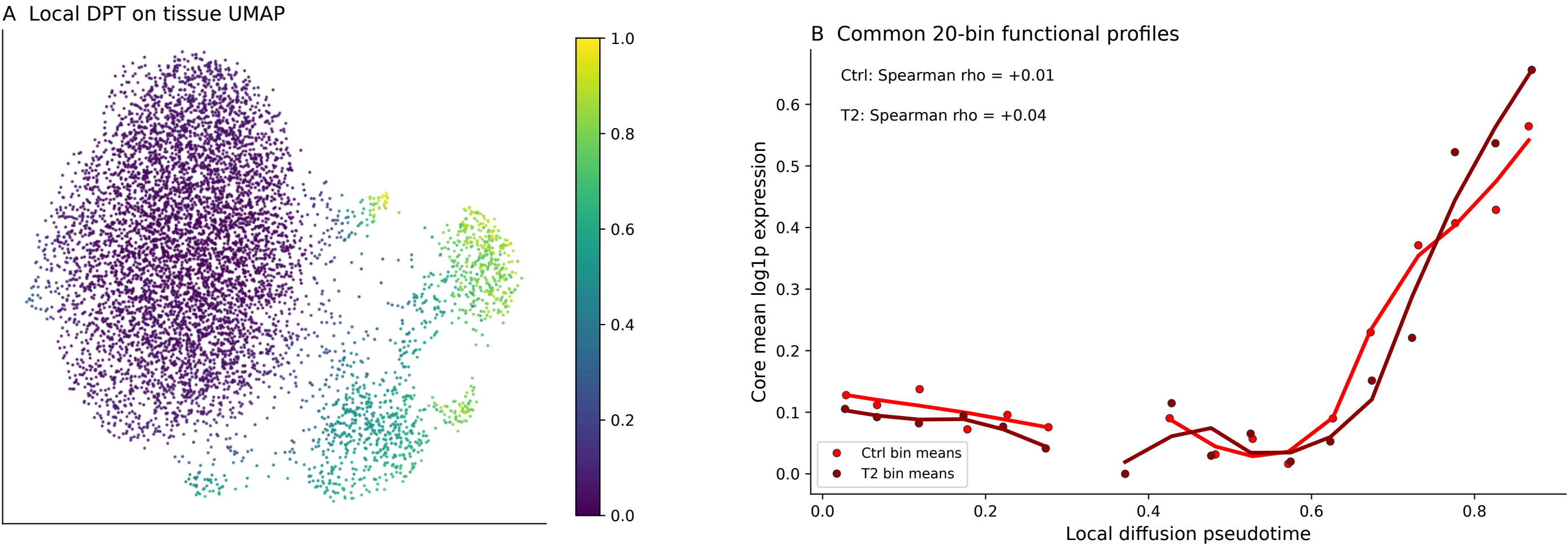


Root state: bone_marrow_S0; measured root markers: Egr2 / Hivep3 (Cd24a unavailable). Root sensitivity rho: 0.89, 0.76.
Analyzed 6,110/6,110 cells; only bins with ≥ 20 cells shown. Smooth lines are descriptive.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Pseudotime is an assumed ordering, not RNA velocity or lineage validation.

Spleen: local cytotoxicity along pseudotime

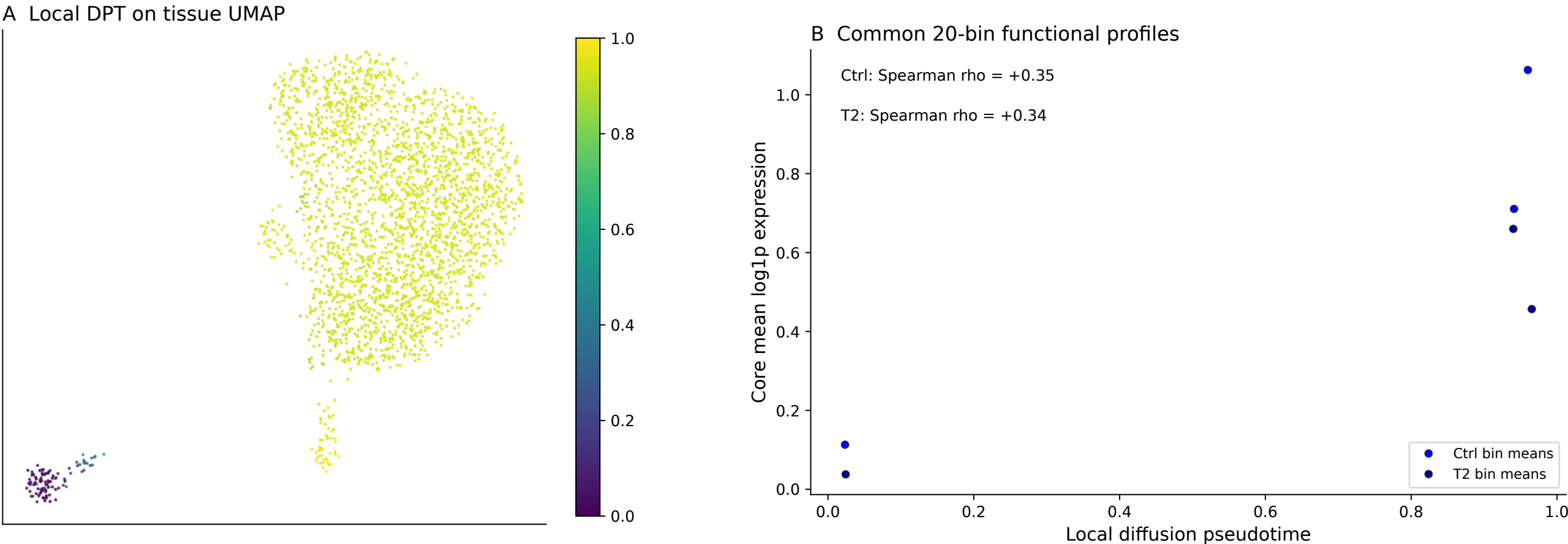
PAGER-style functional curves on a newly computed within-tissue diffusion map.



Root state: spleen_S0; measured root markers: Egr2 / Hivep3 (Cd24a unavailable). Root sensitivity rho: 0.97, 0.76.
Analyzed 6,793/6,793 cells; only bins with ≥ 20 cells shown. Smooth lines are descriptive.

Thymus: local cytotoxicity along pseudotime

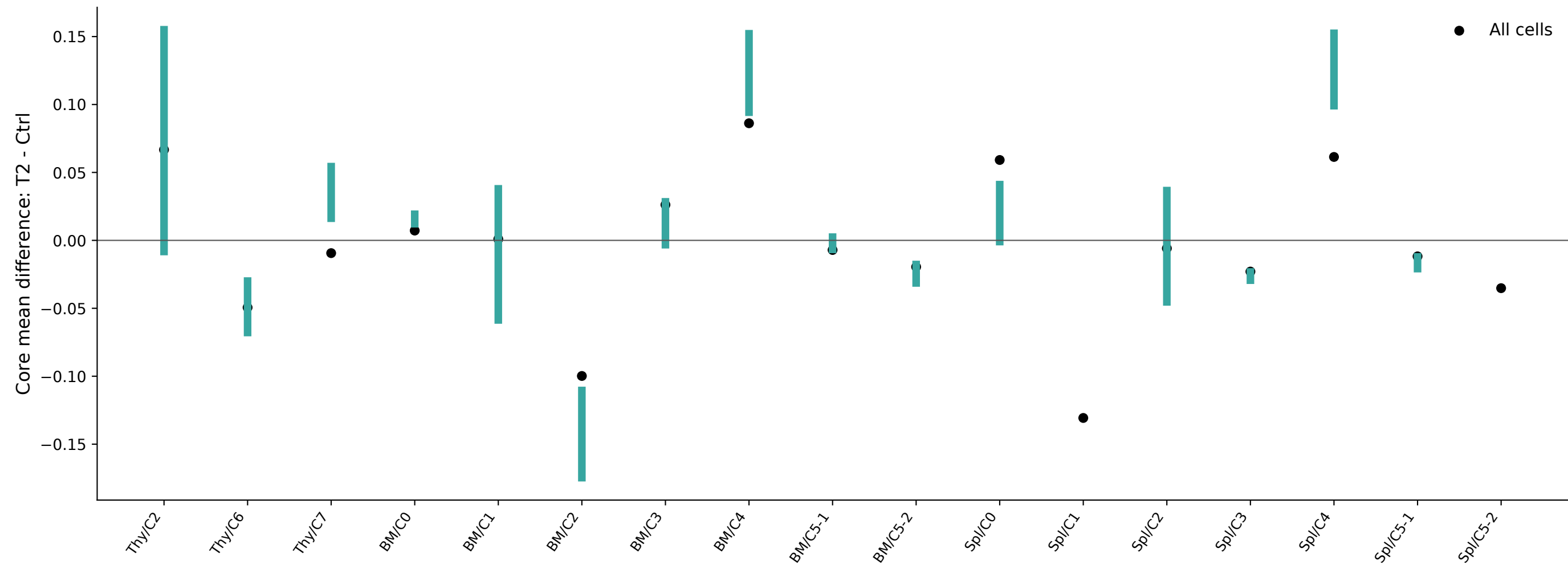
PAGER-style functional curves on a newly computed within-tissue diffusion map.



Root state: thymus_S5; measured root markers: Egr2 / Hivep3 (Cd24a unavailable). Root sensitivity rho: 1.00, 1.00.
Analyzed 2,629/2,629 cells; only bins with ≥ 20 cells shown. Smooth lines are descriptive.

Robustness: QC matching and alternative clustering

Five condition-balanced QC resamples within each tissue x cluster.

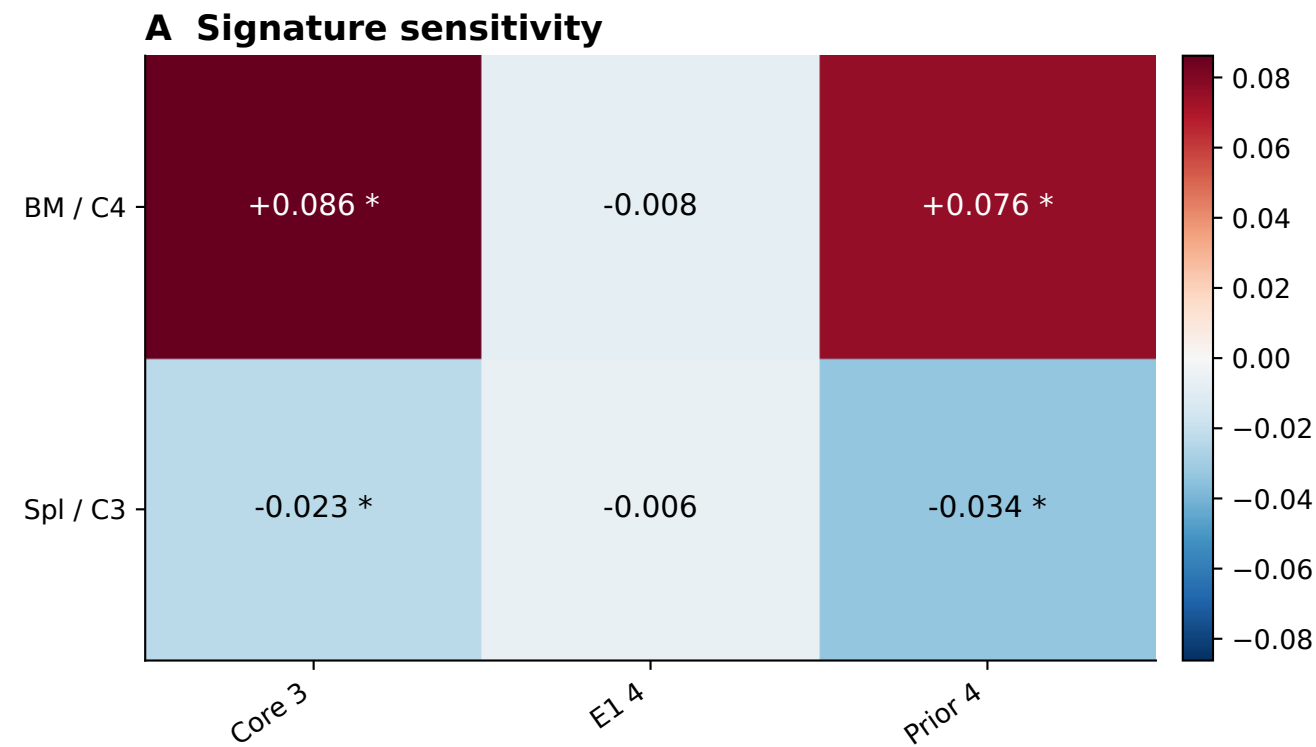


Teal range: five QC-matched effect estimates (not a confidence interval).
Stable tissue clusters are a separate sensitivity analysis; their labels are not interchangeable with original C labels.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Core signature changes do not imply a full killing pathway

Two reproducible local core signals; the broader E1 signature is not significant.



B Complementary evidence from the same dataset

Bone marrow C4 core: +0.086; q = 0.025
Spleen C3 core: -0.023; q = 0.0015
Both: same direction in all 5 QC resamples

Stable-cluster effects: +0.111 and -0.022
Original-cell overlap: 98.7% and 99.5%

E1 (+Ctla2a): q = 0.93 and 0.53
Killing-related GO terms: no q <= 0.05

The evidence supports a localized core-effector transcriptional shift.
It does not establish increased/decreased killing activity or coordinated activation of a full cytotoxic pathway.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Interpretation: localized effector changes with shared metabolic signals

Separate direct observations, working hypotheses and unresolved evidence

Population / scope	Observation	Interpretation
Bone marrow C4	Core score +0.086; q=0.025. Gzmb detection 29% to 44%.	Selective Gzmb-associated effector transcriptional shift.
Spleen C3	Core score -0.023; q=0.0015. Prf1 and Nkg7 decrease.	Small reduction in effector-associated transcripts.
Both populations	T2-up genes enriched for oxidative phosphorylation.	Respiration-related gene expression shifts; no metabolic flux measured.
Limits	Broader E1 signature and killing-related GO are not significant.	No demonstrated change in killing capacity or animal-level treatment effect.

Both core effects persist under QC matching and overlapping stable clusters.
These same-data checks support candidates; independent animals and functional assays remain necessary.